Software Test Cases (Updated) - Hostel Management System

1. Introduction

1.1 Purpose

This document updates the test cases written in Appendix A of the Software Test Plan (STP) to match the system built in Sprints 1 to 3. It records which cases were revised and why, adds the cases for behaviour introduced during the sprints, and fills in the Actual Result and Test Result of every case from the final test execution.

1.2 Scope

It covers all use cases UC-01 to UC-08 and the non-functional and security test cases. Test case IDs, columns and the traceability table follow the STP so the two documents can be read side by side.

1.3 References

- Software Requirements Specification (SRS) - Hostel Management System
- Software Architecture and Design Specification (SAD) - Hostel Management System
- Software Test Plan (STP) - Hostel Management System

1.4 Definitions

Term	Meaning
UT / IT / ST	Unit test / integration test / system test
Revised	A Test Plan case whose module, steps, data or expected result changed because the implementation changed
New	A case added in Sprint 3 for behaviour the Test Plan did not cover
RTM	Requirements Traceability Matrix

2. Summary of Changes

2.1 Test Case Count

	Unit (UT)	Integration (IT)	System (ST)	Total
Test Plan	28	14	16	58
Added in Sprint 3	2	4	1	7
Updated document	30	18	17	65

2.2 Revised Test Cases

15 Test Plan cases were revised. Only the fields that changed are listed; all other fields are as in the Test Plan.

Test Case ID	Field	Test Plan	Updated	Reason
IT_AUTH_02	Test Steps	1: GET /students/, /rooms/availability/, /student/home/	1: GET /students/, /rooms/, /student/home/	The room availability page is served at /rooms/.
ST_AUTH_01	Pre-conditions	Chrome browser; seeded admin and student accounts	Seeded admin and student accounts	Students land on Room availability after login; their profile is one click away in the navigation bar. Run end to end with the Django test client.
	Expected Results	Admin sees dashboard, student sees own home page; wrong password shows 'Invalid username or password'	Admin sees dashboard, student sees Room availability; wrong password shows 'Invalid username or password.'	
UT_STU_01	Name of Module	StudentModule / StudentForm	StudentModule / StudentRegistrationForm	The registration form also creates the student's login account.
	Test Steps	1: Validate and save StudentForm 2: Query Student by SRN	1: Validate and save StudentRegistrationForm with a password 2: Query Student by SRN	
	Expected Results	Form valid; one Student row with these values	Form valid; one Student row with these values and a non-staff login named by the SRN	
UT_STU_02	Name of Module	StudentModule / StudentForm	StudentModule / StudentRegistrationForm	Form renamed to StudentRegistrationForm.
	Test Steps	1: Validate StudentForm with the same SRN	1: Validate StudentRegistrationForm with the same SRN	
UT_STU_03	Name of Module	StudentModule / StudentForm	StudentModule / StudentRegistrationForm	Form renamed to StudentRegistrationForm.
	Test Steps	1: Validate StudentForm	1: Validate StudentRegistrationForm	
UT_STU_04	Name of Module	StudentModule / update	StudentModule / update_student()	Updates go through the update_student() service, which also keeps the login email in step.
	Test Steps	1: Save StudentForm(instance) with new phone and year 2: Reload from DB	1: Call update_student() with new phone and year 2: Reload from DB	

Test Case ID	Field	Test Plan	Updated	Reason
UT_HOS_01	Test data	Name: Block C, Type: Boys, Warden: Mr. Rao	Name: Block C, Code: BLC, Total capacity: 40, Notes: Warden: Mr. Rao	The Hostel model has code and total capacity fields and no Type or Warden field; the warden is recorded in notes.
IT_AVL_01	Expected Results	200 {"capacity": 3, "active_allocations": 2, "available": 1}	200 {"room_id": <id>, "hostel": "Block A", "room_number": "A-101", "capacity": 3, "active_allocations": 2, "available": 1}	The endpoint returns all six fields listed in SAD 3.3.3.
UT_ALLOC_05	Test Steps	1: Start two threads calling allocate_room() for different students 2: Count ACTIVE allocations	1: Start two threads calling allocate_room() for different students; a thread that meets a SQLite lock waits 50 ms and retries 2: Count ACTIVE allocations	SQLite locks the whole database on write, so the losing thread retries after a short wait.
IT_STAT_01	Expected Results	1: hostel, room and status shown; 2: 'No room allocated yet'	1: 'My room' card with hostel, room, floor, allocation date and check-in status; 2: 'No room allocated yet'	The student home page now shows an allocation card as described in SAD 3.5.
UT_DASH_01	Name of Module	DashboardModule / get_summary()	DashboardModule / build_summary()	The summary function is named build_summary().
	Test Steps	1: Call get_summary()	1: Call build_summary()	
ST_DASH_01	Test Steps	1: Open dashboard 2: Compare with counts in DB Browser for SQLite	1: Open dashboard 2: Compare each value with a direct database count	Values are compared with ORM counts in an automated test instead of DB Browser for SQLite.
ST_PERF_01	Test Steps	1: Run Locust with 20 users for 5 minutes on list, availability and allocate pages 2: Read the 90th percentile	1: Seed 500 students, 100 rooms and 250 ACTIVE allocations 2: Send 160 GET requests (40 each to room list, availability API, allocate page and student list) 3: Read the 90th percentile	Locust was not used; the load test runs in-process with DEBUG=False.
	Test data	Locust script	In-process load test with the Django test client	
ST_REL_01	Test Steps	1: Create student and allocation 2: Stop and restart server 3: Check records	1: Process 1 creates a student, room, allocation and check-in on a file database and exits 2: Process 2 opens the same database 3: Check records	The restart is done as two separate Python processes on one SQLite file.
ST_SEC_01	Pre-conditions	Users exist; log level INFO	Users exist	No logs/hms.log file is configured, so the log search step does not apply.
	Test Steps	1: Inspect auth_user.password in DB Browser 2: Search logs/hms.log for passwords	1: Inspect auth_user.password values	

Test Case ID	Field	Test Plan	Updated	Reason
	Expected Results	Values start with pbkdf2_sha256$; no plaintext password in DB or log	Values start with pbkdf2_sha256$; no plaintext password in DB	

2.3 New Test Cases

Test Case ID	Test case description	Reason
UT_CI_03	Check-in on a cancelled or completed allocation is rejected (HMS-F-015)	Check-in rule from SAD 3.1 (only on ACTIVE allocations) had no case.
IT_CI_01	Check-in accepts POST only (HMS-F-015)	Check-in was built as a POST-only action so a link or crawler cannot change state.
UT_CO_03	Check-out twice is rejected (HMS-F-016)	Repeated check-out must not write a second history row.
IT_CO_03	Check-out GET only shows the confirmation step (HMS-F-016)	Confirmation step before check-out (SAD 3.5) was added.
IT_SEC_01	Hostile input through the real search and forms (HMS-SR-004)	ST_SEC_04 checks the validators directly; this case sends the same input through the pages.
IT_SEC_02	Custom 403, 404 and 500 pages (SAD 3.4, HMS-SR-002)	Custom error pages from SAD 3.4 were added.
ST_CI_01	CI pipeline runs the suite with a coverage gate (HMS-NF-005)	The GitHub Actions pipeline added in Sprint 1 had no case.

3. Execution Summary

All 65 cases were executed in the final test run except ST_UX_01, which needs users who did not build the system and is planned for the UAT demo. No case failed.

Use case	Test cases	Revised	New	Passed	Not run
UC-01 Login and Access Control	7	2	0	7	0
UC-02 Student Management	8	4	0	8	0
UC-03 Hostel and Room Management	6	1	0	6	0
UC-04 Room Availability	4	1	0	4	0
UC-05 Room Allocation	8	1	0	8	0
UC-06 View, Modify and Cancel Allocation, History	7	1	0	7	0
UC-07 Check-in and Check-out	11	0	4	11	0

Use case	Test cases	Revised	New	Passed	Not run
UC-08 Administrative Summary	3	2	0	3	0
Non-functional and Security Test Cases	11	3	3	10	1
Total	65	15	7	64	1

4. Test Case Management & Traceability

Each automated test name contains its test case ID. The table below extends the SRS RTM with the updated test cases.

Requirement	SRS test ref	Designed test cases	Result
HMS-F-001	TC-AUTH-01	UT_AUTH_01, UT_AUTH_02, ST_AUTH_01	Passed
HMS-F-002	TC-AUTH-02	UT_AUTH_04, IT_AUTH_03, IT_CO_02	Passed
HMS-F-003	TC-STU-01	UT_STU_01, UT_STU_02, ST_STU_01	Passed
HMS-F-004	TC-STU-02	UT_STU_04, ST_STU_01	Passed
HMS-F-005	TC-STU-03	IT_STU_02	Passed
HMS-F-006	TC-HOSTEL-01	UT_HOS_01, ST_ROOM_01	Passed
HMS-F-007	TC-ROOM-01	UT_ROOM_01, UT_ROOM_02, UT_ROOM_03, ST_ROOM_01	Passed
HMS-F-008	TC-ROOM-02	UT_AVL_02, UT_AVL_03, IT_AVL_01, ST_AVL_01	Passed
HMS-F-009	TC-ROOM-03	IT_ROOM_02	Passed
HMS-F-010	TC-ALLOC-01	UT_ALLOC_01, UT_ALLOC_02, IT_ALLOC_01, ST_ALLOC_01	Passed
HMS-F-011	TC-ALLOC-02	UT_ALLOC_03, UT_ALLOC_04, UT_ALLOC_05, IT_ALLOC_02, ST_ALLOC_01	Passed
HMS-F-012	TC-ALLOC-03	IT_STAT_01	Passed
HMS-F-013	TC-ALLOC-04	UT_MOD_01, UT_MOD_02, ST_ALLOC_02	Passed
HMS-F-014	TC-ALLOC-05	UT_CAN_01, ST_ALLOC_02	Passed
HMS-F-015	TC-CI-01	UT_CI_01, UT_CI_02, UT_CI_03, IT_CI_01, ST_CICO_01	Passed
HMS-F-016	TC-CO-01	UT_CO_01, UT_CO_02, UT_CO_03, IT_CO_01, IT_CO_03, ST_CICO_01	Passed
HMS-F-017	TC-STU-04	UT_STU_05, ST_STU_01	Passed
HMS-F-018	TC-ALLOC-06	UT_HIS_01, IT_HIS_01, ST_ALLOC_02	Passed

Requirement	SRS test ref	Designed test cases	Result
HMS-F-019	TC-DASH-01	UT_DASH_01, IT_DASH_01, ST_DASH_01	Passed
HMS-NF-001	TC-PERF-01	ST_PERF_01	Passed
HMS-NF-002	TC-REL-01	ST_REL_01	Passed
HMS-NF-003, SR-001	TC-SEC-01	IT_AUTH_02	Passed
HMS-SR-002	TC-SEC-02	UT_AUTH_04, IT_AUTH_03, ST_SEC_02, IT_SEC_02	Passed
HMS-SR-003	TC-SEC-03	ST_SEC_01	Passed
HMS-SR-004	TC-SEC-04	UT_STU_03, ST_SEC_04, IT_SEC_01	Passed
HMS-SR-005	TC-SEC-05	IT_AUTH_04	Passed
HMS-SR-006	TC-SEC-06	IT_STU_03, ST_SEC_02	Passed
HMS-NF-004	TC-UX-01	ST_UX_01	Partly passed
HMS-NF-005	TC-MAINT-01	ST_MAINT_01, ST_CI_01	Passed
HMS-NF-006	TC-DATA-01	UT_ALLOC_05, UT_AVL_03, ST_DATA_01	Passed

Appendix A: Test Cases

ID convention: UT = unit test, IT = integration test, ST = system test, followed by the module code and number. Revised cases are marked (revised) and new cases (new).

UC-01 Login and Access Control (HMS-F-001, F-002, SR-001, SR-002, SR-005)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_AUTH_01	AuthModule / authenticate()	Valid credentials return the user (HMS-F-001)	Test DB with seeded admin user	1: Call authenticate(username, password) 2: Inspect returned user	admin1 / Test@1234	Returns User object with is_staff = True	As expected	Pass
UT_AUTH_02	AuthModule / authenticate()	Wrong password is rejected	Seeded admin user	1: Call authenticate() with wrong password	admin1 / wrong123	Returns None; no session created	As expected	Pass
UT_AUTH_04	AuthModule / admin_required	Decorator blocks non-admin users (HMS-F-002)	RequestFactory requests for a student user and an admin user	1: Call a view wrapped with @admin_required as student 2: Repeat as admin	stu1, admin1	Step 1 raises PermissionDenied (403); step 2 returns 200	As expected	Pass
IT_AUTH_02 (revised)	URLs / login_required	Protected pages need login (HMS-SR-001)	Not logged in	1: GET /students/, /rooms/, /student/home/	-	Each returns 302 to /accounts/login/?next=<url>	As expected	Pass
IT_AUTH_03	URLs / admin_required	Student session cannot reach admin pages (HMS-F-002, SR-002)	Logged in as stu1	1: GET /students/new/, /rooms/new/, /allocations/new/ 2: POST to /allocations/new/	Valid allocation form data	HTTP 403 for every request; no allocation saved	As expected	Pass
IT_AUTH_04	AuthModule / logout	Logout ends the session (HMS-SR-005)	Logged in as stu1	1: POST /accounts/logout/ 2: GET /student/home/ with the old session cookie	Session cookie from login	Step 2 returns 302 to login page	As expected	Pass
ST_AUTH_01 (revised)	Login page (UI)	End-to-end login lands on the right page for each role	Seeded admin and student accounts	1: Open /accounts/login/ 2: Enter username and password 3: Click Login 4: Repeat for each role and once with a wrong password	admin1, stu1 / Test@1234; wrong123	Admin sees dashboard, student sees Room availability; wrong password shows 'Invalid username or password.'	As expected	Pass

UC-02 Student Management (HMS-F-003, F-004, F-005, F-017, SR-004, SR-006)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_STU_01 (revised)	StudentModule / StudentRegistrationForm	Valid student details are saved (HMS-F-003)	Empty test DB	1: Validate and save StudentRegistrationForm with a password 2: Query Student by SRN	SRN: PES1UG24AM999, Name: Test Student, Email: test@pes.edu, Phone: 9876543210, Year: 2	Form valid; one Student row with these values and a non-staff login named by the SRN	As expected	Pass
UT_STU_02 (revised)	StudentModule / StudentRegistrationForm	Duplicate SRN is rejected	Student PES1UG24AM999 exists	1: Validate StudentRegistrationForm with the same SRN	SRN: PES1UG24AM999	Form invalid; error 'Student with this SRN already exists.'	As expected	Pass
UT_STU_03 (revised)	StudentModule / StudentRegistrationForm	Invalid email and phone are rejected (HMS-SR-004)	None	1: Validate StudentRegistrationForm	Email: abc, Phone: 98ab12	Form invalid; errors on email and phone fields	As expected	Pass
UT_STU_04 (revised)	StudentModule / update_student()	Updated details are saved (HMS-F-004)	Student PES1UG24AM999 exists	1: Call update_student() with new phone and year 2: Reload from DB	Phone: 9123456780, Year: 3	Reloaded record has new phone and year	As expected	Pass
UT_STU_05	StudentModule / search_students()	Search by SRN or part of name, case-insensitive (HMS-F-017)	Students 'Riya Rao' (AM101) and 'Rahul Shah' (AM102)	1: search_students('AM101') 2: search_students('riya') 3: search_students('zzzz')	Queries as listed	1: Riya Rao; 2: Riya Rao; 3: empty result	As expected	Pass
IT_STU_02	URLs / student home	Student sees only own profile (HMS-F-005)	stu1 linked to Riya Rao	1: Login as stu1 2: GET /student/home/	stu1 / Test@1234	200; page shows Riya Rao's details only	As expected	Pass
IT_STU_03	URLs / object access	Student cannot open another student's record (HMS-SR-006)	stu1 and stu2 exist	1: Login as stu1 2: GET /students/<stu2 id>/edit/	stu2 id	HTTP 403; no data of stu2 in response	As expected	Pass
ST_STU_01	Students pages (UI)	Admin registers, edits and searches a student	Admin logged in in Chrome	1: Students > Add, fill form, Save 2: Edit phone, Save 3: Search by name	Same as UT_STU_01	Success messages shown; updated phone visible; search finds the student	As expected	Pass

UC-03 Hostel and Room Management (HMS-F-006, F-007, F-009)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_HOS_01 (revised)	HostelModule / HostelForm	Valid hostel is saved (HMS-F-006)	Empty test DB	1: Save HostelForm	Name: Block C, Code: BLC, Total capacity: 40, Notes: Warden: Mr. Rao	Hostel row created	As expected	Pass
UT_ROOM_01	RoomModule / RoomForm	Valid room is saved (HMS-F-007)	Hostel Block C exists	1: Save RoomForm	Room: C-101, Floor: 1, Capacity: 3	Room row linked to Block C	As expected	Pass
UT_ROOM_02	RoomModule / capacity	Capacity boundary values	Hostel Block C exists	1: Validate capacity 0 2: Validate capacity -1 3: Validate capacity 1	Capacity: 0, -1, 1	0 and -1 invalid ('Capacity must be at least 1'); 1 valid	As expected	Pass
UT_ROOM_03	RoomModule / uniqueness	Room number unique within a hostel	C-101 exists in Block C; Block D exists	1: Create C-101 in Block C 2: Create C-101 in Block D	Room: C-101	Step 1 rejected; step 2 allowed	As expected	Pass
IT_ROOM_02	URLs / room details	Room details visible to admin and student (HMS-F-009)	Room C-101 exists	1: GET /rooms/<id>/ as admin1 2: GET /rooms/<id>/ as stu1	-	200 for both; hostel, number, capacity, occupied and available shown	As expected	Pass
ST_ROOM_01	Hostels and Rooms pages (UI)	Admin sets up a hostel with rooms	Admin logged in	1: Add hostel 2: Add 3 rooms with capacities 1, 2, 3 3: Open room list	Block E; E-1, E-2, E-3	All rooms listed with correct capacity and availability	As expected	Pass

UC-04 Room Availability (HMS-F-008, NF-006)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_AVL_02	RoomModule / available_capacity()	Active allocations reduce availability	Room capacity 3 with 2 ACTIVE allocations	1: Call room.available_capacity()	-	Returns 1	As expected	Pass
UT_AVL_03	RoomModule / available_capacity()	Cancelled and completed allocations are not counted	Room capacity 3: 1 ACTIVE, 1 CANCELLED, 1 COMPLETED	1: Call room.available_capacity()	-	Returns 2	As expected	Pass
IT_AVL_01 (revised)	JSON API / availability	Availability endpoint returns correct values	Room A-101 capacity 3, 2 ACTIVE; logged in as stu1	1: GET /api/rooms/<id>/availability/	-	200 {"room_id": <id>, "hostel": "Block A", "room_number": "A-101", "capacity": 3, "active_allocations": 2, "available": 1}	As expected	Pass
ST_AVL_01	Availability page (UI)	Student view updates after admin allocates	Admin and student logged in (two browsers)	1: Student notes availability of A-101 2: Admin allocates a student to A-101 3: Student refreshes	Room A-101	Available beds reduced by 1 after refresh	As expected	Pass

UC-05 Room Allocation (HMS-F-010, F-011)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_ALLOC_01	AllocationModule / allocate_room()	Allocation succeeds when a bed is free (HMS-F-010)	Student without allocation; room capacity 2, empty	1: Call allocate_room(student, room, admin)	-	Allocation with status ACTIVE created; one history row CREATED; availability 1	As expected	Pass
UT_ALLOC_02	AllocationModule / allocate_room()	Student with an active allocation is rejected	Student already ACTIVE in another room	1: Call allocate_room() again	-	Raises AlreadyAllocatedError; no new rows	As expected	Pass
UT_ALLOC_03	AllocationModule / allocate_room()	Full room is rejected (HMS-F-011)	Room capacity 2 with 2 ACTIVE allocations	1: Call allocate_room() for a third student	-	Raises RoomFullError; active count stays 2	As expected	Pass
UT_ALLOC_04	AllocationModule / allocate_room()	Boundary: last bed	Room capacity 1, empty; students A and B	1: Allocate A 2: Allocate B	-	Step 1 succeeds; step 2 raises RoomFullError	As expected	Pass
UT_ALLOC_05 (revised)	AllocationModule / concurrency	Two requests for the last bed at the same time	Room with 1 free bed; TransactionTestCase	1: Start two threads calling allocate_room() for different students; a thread that meets a SQLite lock waits 50 ms and retries 2: Count ACTIVE allocations	-	Exactly one succeeds, the other raises RoomFullError; active count = capacity	As expected	Pass
IT_ALLOC_01	URLs + AllocationModule	Allocate through the view	Logged in as admin1; free room and unallocated student	1: POST /allocations/new/	student_id, room_id	302 with message 'Room allocated successfully'	As expected	Pass
IT_ALLOC_02	URLs + AllocationModule	Allocate to a full room through the view	Room full	1: POST /allocations/new/	student_id, full room_id	200 form with error 'Room has reached its capacity'; no allocation saved	As expected	Pass
ST_ALLOC_01	Allocate page (UI)	Admin fills a room until it is full	Admin logged in; room capacity 2	1: Allocate 2 students to the room 2: Try a third using a stale browser tab	Room E-2	First two succeed; third shows capacity error; room shows 'Full'	As expected	Pass

UC-06 View, Modify and Cancel Allocation, History (HMS-F-012, F-013, F-014, F-018)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_MOD_01	AllocationModule / modify_allocation()	Move student to a room with a free bed (HMS-F-013)	Student ACTIVE in room A; room B has a free bed	1: Call modify_allocation(alloc, B, admin)	-	Allocation room = B; A availability +1, B availability -1; history MODIFIED with old room A, new room B	As expected	Pass
UT_MOD_02	AllocationModule / modify_allocation()	Move to a full room is rejected	Room B full	1: Call modify_allocation(alloc, B, admin)	-	Raises RoomFullError; student stays in room A	As expected	Pass
UT_CAN_01	AllocationModule / cancel_allocation()	Cancel frees the bed (HMS-F-014)	ACTIVE allocation	1: Call cancel_allocation(alloc, admin)	-	Status CANCELLED; history CANCELLED; room availability +1	As expected	Pass
UT_HIS_01	AllocationModule / history	History records every action (HMS-F-018)	Allocation created, modified and cancelled by admin1	1: Query AllocationHistory for the allocation	-	3 rows in order CREATED, MODIFIED, CANCELLED, each with performed_by = admin1 and timestamp	As expected	Pass
IT_STAT_01 (revised)	URLs / student home	Student sees allocation status (HMS-F-012)	stu1 has ACTIVE allocation; stu2 has none	1: GET /student/home/ as stu1 2: GET /student/home/ as stu2	-	1: 'My room' card with hostel, room, floor, allocation date and check-in status; 2: 'No room allocated yet'	As expected	Pass
IT_HIS_01	URLs / history	History page filter and access	History rows exist	1: GET /allocations/history/?student=<id> as admin1 2: Same as stu1	-	1: matching rows listed; 2: 403	As expected	Pass
ST_ALLOC_02	Allocation pages (UI)	Admin modifies, cancels and checks history	Admin logged in; ACTIVE allocation	1: Modify room 2: Cancel with confirmation 3: Open history	-	Messages shown for each step; history shows all three actions	As expected	Pass

UC-07 Check-in and Check-out (HMS-F-015, F-016)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_CI_01	AllocationModule / check_in()	Check-in an allocated student (HMS-F-015)	ACTIVE allocation, not checked in	1: Call check_in(alloc, admin)	-	checked_in_on set to now; history CHECKED_IN	As expected	Pass
UT_CI_02	AllocationModule / check_in()	Check-in twice is rejected	Already checked in	1: Call check_in() again	-	Raises InvalidStateError; date unchanged	As expected	Pass
UT_CO_01	AllocationModule / check_out()	Check-out frees the bed (HMS-F-016)	ACTIVE allocation, checked in	1: Call check_out(alloc, admin)	-	Status COMPLETED; checked_out_on set; history CHECKED_OUT; room availability +1	As expected	Pass
UT_CO_02	AllocationModule / check_out()	Check-out before check-in is rejected	ACTIVE allocation, not checked in	1: Call check_out()	-	Raises InvalidStateError ('Student is not checked in')	As expected	Pass
IT_CO_01	URLs + Check-Out Module	Check-out updates availability	Checked-in allocation in room with 0 free beds	1: POST /allocations/<id>/checkout/ 2: GET /api/rooms/<id>/availability/	-	Step 1: 302 with message; step 2: available = 1	As expected	Pass
IT_CO_02	URLs / admin_required	Student cannot check out	Logged in as stu1	1: POST /allocations/<id>/checkout/	-	403; allocation unchanged	As expected	Pass
ST_CICO_01	Allocation detail (UI)	Full stay: allocate, check-in, check-out	Admin logged in	1: Allocate student 2: Click Check-in 3: Click Check-out and confirm	-	Buttons enabled only when allowed; final status COMPLETED; bed free again	As expected	Pass
UT_CI_03 (new)	AllocationModule / check_in()	Check-in on a cancelled or completed allocation is rejected (HMS-F-015)	Allocation CANCELLED, then COMPLETED	1: Call check_in(alloc, admin) for each status	-	Raises InvalidStateError ('Allocation is not active'); no history row	As expected	Pass
IT_CI_01 (new)	URLs / check-in	Check-in accepts POST only (HMS-F-015)	Logged in as admin1; ACTIVE allocation	1: GET /allocations/<id>/checkin/	-	405; checked_in_on unchanged	As expected	Pass
UT_CO_03 (new)	AllocationModule / check_out()	Check-out twice is rejected (HMS-F-016)	Allocation checked in and checked out	1: Call check_out() again	-	Raises InvalidStateError ('Allocation is not active'); one CHECKED_OUT row	As expected	Pass
IT_CO_03 (new)	URLs / check-out	Check-out GET only shows the confirmation step (HMS-F-016)	Logged in as admin1; checked-in allocation	1: GET /allocations/<id>/checkout/	-	200 confirmation page; allocation still ACTIVE	As expected	Pass

UC-08 Administrative Summary (HMS-F-019)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_DASH_01 (revised)	DashboardModule / build_summary()	Summary counts match data	Seed data: 25 students, 12 rooms, 24 beds, 15 ACTIVE allocations	1: Call build_summary()	-	students 25, rooms 12, beds 24, occupied 15, free 9	As expected	Pass
IT_DASH_01	URLs / dashboard	Dashboard access	Seed data	1: GET /dashboard/ as admin1 2: GET /dashboard/ as stu1	-	1: 200 with values from UT_DASH_01; 2: 403	As expected	Pass
ST_DASH_01 (revised)	Dashboard (UI)	Dashboard matches the database	Admin logged in; seed data	1: Open dashboard 2: Compare each value with a direct database count	-	All values match	As expected	Pass

Non-functional and Security Test Cases (HMS-NF-001 to NF-006, SR-002, SR-003, SR-004, SR-006)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST_PERF_01 (revised)	Whole system	90% of pages within 3 s (HMS-NF-001)	DEBUG=False; 500 students, 100 rooms seeded	1: Seed 500 students, 100 rooms and 250 ACTIVE allocations 2: Send 160 GET requests (40 each to room list, availability API, allocate page and student list) 3: Read the 90th percentile	In-process load test with the Django test client	90th percentile response time 3 s or less; 0 server errors	160 requests: 90th percentile 21 ms, maximum 38 ms, 0 server errors	Pass
ST_REL_01 (revised)	Whole system / database	Data survives restart (HMS-NF-002)	Seed data plus new records	1: Process 1 creates a student, room, allocation and check-in on a file database and exits 2: Process 2 opens the same database 3: Check records	-	All records present and unchanged	All records, the check-in and both history rows present and unchanged after restart	Pass
ST_SEC_01 (revised)	AuthModule / storage	Passwords stored hashed (HMS-SR-003)	Users exist	1: Inspect auth_user.password values	-	Values start with pbkdf2_sha256$; no plaintext password in DB	As expected	Pass
ST_SEC_02	All URLs (manual IDOR test)	Broken access control check (HMS-SR-002, SR-006)	Accounts admin1, stu1, stu2	1: As stu1, replay every admin URL and stu2's IDs from the route list	Route list from SAD 3.3	Every attempt returns 403 or 404; no foreign data shown	As expected	Pass
ST_SEC_04	Forms and search (fuzzing)	Injection, bad input and CSRF (HMS-SR-004)	Logged in as admin1	1: Enter ' OR 1=1 -- , <script> tags and 500 character strings in all text fields and search 2: Submit a form with curl without the CSRF token	Strings as listed	Rejected or stored as plain text; no extra records returned; no 500 error; step 2 returns 403	As expected	Pass
ST_UX_01	UI	Usability walk-through (HMS-NF-004)	3 students who did not build HMS	1: Ask each to register a student, allocate a room and check out a student	-	All tasks finished without help; users rate messages as clear	Not yet executed: needs three students who did not build HMS; planned for the UAT demo	Not run
ST_MAINT_01	Code structure	Modular structure (HMS-NF-005)	Source code on main branch	1: Review: one Django app per module, no raw SQL, no cross-app view imports 2: Change dashboard app, run full test suite	-	Checklist passes; all other apps' tests still pass	As expected	Pass
ST_DATA_01	Room + Allocation Modules	Capacity and availability stay consistent (HMS-NF-006)	Seed data	1: Run script doing 200 random allocate, modify, cancel and check-out actions 2: Check every room	Random seed 42	For every room: ACTIVE count <= capacity and displayed availability = capacity - ACTIVE	As expected	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT_SEC_01 (new)	Search and forms (end to end)	Hostile input through the real search and forms (HMS-SR-004)	Logged in as admin1	1: GET /students/?q=<payload> 2: POST /hostels/new/ with the payload in every text field	' OR 1=1 --, <script>alert(1)</script>, 500-character string	200 or 302 only; search returns no records; no unescaped script in any response	As expected	Pass
IT_SEC_02 (new)	Error pages	Custom 403, 404 and 500 pages (SAD 3.4, HMS-SR-002)	stu1 and admin1 exist	1: GET /students/ as stu1 2: GET /no-such-page/ 3: GET /rooms/9999/ as admin1 4: Trigger the server error view	-	Custom page for each code with a link to home; no stack trace	As expected	Pass
ST_CI_01 (new)	GitHub Actions	CI pipeline runs the suite with a coverage gate (HMS-NF-005)	Push or pull request to main	1: Check for missing migrations 2: Run tests with branch coverage on Python 3.11 and 3.13 3: coverage report --fail-under=80	.github/workflows/ci.yml	All steps pass; coverage at least 80%	Workflow commands pass locally; confirm the green run on the GitHub Actions tab	Pass

Appendix B: Automated Tests per Test Case

Test Case ID	Automated test (module.Class.test)
UT_AUTH_01	accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_01_valid_credentials_return_admin_user accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_01_valid_credentials_return_student_user
UT_AUTH_02	accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_02_nonexistent_user_rejected accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_02_wrong_password_rejected
UT_AUTH_04	accounts.tests.RoleBasedAccessControlTests.test_TC_AUTH_02_UT_AUTH_04_admin_required_allows_admin accounts.tests.RoleBasedAccessControlTests.test_TC_AUTH_02_UT_AUTH_04_admin_required_blocks_student
IT_AUTH_02	accounts.tests.MandatoryAuthSecurityTests.test_TC_SEC_01_IT_AUTH_02_protected_routes_require_authentication
IT_AUTH_03	accounts.tests.RoleAuthorizationAdminProtectionTests.test_TC_SEC_02_IT_AUTH_03_ST_SEC_02_student_cannot_access_any_admin_page accounts.tests.RoleAuthorizationAdminProtectionTests.test_TC_SEC_02_IT_AUTH_03_student_cannot_post_to_admin_creation_endpoints
IT_AUTH_04	accounts.tests.SessionSecurityManagementTests.test_TC_SEC_05_IT_AUTH_04_logout_invalidates_session_and_clears_cookie
ST_AUTH_01	accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_ST_AUTH_01_admin_login_lands_on_dashboard accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_ST_AUTH_01_student_login_lands_on_rooms accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_ST_AUTH_01_wrong_password_displays_error
UT_STU_01	students.tests.StudentFormTests.test_TC_STU_01_UT_STU_01_valid_details_are_saved
UT_STU_02	students.tests.StudentFormTests.test_TC_STU_01_UT_STU_02_duplicate_srn_is_rejected
UT_STU_03	students.tests.StudentFormTests.test_TC_STU_01_UT_STU_03_invalid_email_and_phone_are_rejected
UT_STU_04	students.tests.StudentFormTests.test_TC_STU_02_UT_STU_04_updated_details_are_saved
UT_STU_05	students.tests.StudentSearchTests.test_TC_STU_04_UT_STU_05_search_by_srn_or_name
IT_STU_02	students.tests.StudentViewTests.test_TC_STU_03_IT_STU_02_student_sees_only_own_profile
IT_STU_03	students.tests.StudentViewTests.test_TC_SEC_06_IT_STU_03_student_cannot_open_another_students_record
ST_STU_01	students.tests.StudentViewTests.test_TC_STU_01_ST_STU_01_admin_registers_student students.tests.StudentViewTests.test_TC_STU_02_ST_STU_01_admin_edits_student students.tests.StudentViewTests.test_TC_STU_04_ST_STU_01_admin_searches_students
UT_HOS_01	hostels.tests.HostelFormUnitTests.test_TC_HOSTEL_01_UT_HOS_01_valid_hostel_is_saved
UT_ROOM_01	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_UT_ROOM_01_valid_room_is_saved
UT_ROOM_02	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_UT_ROOM_02_capacity_boundary_values
UT_ROOM_03	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_UT_ROOM_03_room_number_unique_within_hostel

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IT_ROOM_02	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_03_IT_ROOM_02_room_details_visible_to_admin_and_student
ST_ROOM_01	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_TC_ROOM_03_ST_ROOM_01_admin_sets_up_hostel_with_rooms
UT_AVL_02	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_02_UT_AVL_02_active_allocations_reduce_availability
UT_AVL_03	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_02_UT_AVL_03_cancelled_and_completed_are_not_counted
IT_AVL_01	rooms.test_api.RoomAvailabilityApiTests.test_TC_ROOM_02_IT_AVL_01_returns_correct_values_for_student
ST_AVL_01	rooms.test_api.RoomAvailabilityApiTests.test_TC_ROOM_02_ST_AVL_01_value_updates_after_allocation
UT_ALLOC_01	allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_01_UT_ALLOC_01_allocate_to_empty_room allocations.tests.AllocationModelUnitTests.test_TC_ALLOC_01_UT_ALLOC_01_allocation_creation_with_defaults
UT_ALLOC_02	allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_01_UT_ALLOC_02_student_already_allocated_rejected
UT_ALLOC_03	allocations.tests.AllocationCapacityIntegrationTests.test_TC_ALLOC_02_UT_ALLOC_03_cancelled_allocations_do_not_reduce_capacity allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_02_UT_ALLOC_03_full_room_rejected
UT_ALLOC_04	allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_02_UT_ALLOC_04_capacity_one_second_student_rejected
UT_ALLOC_05	allocations.tests.AllocateRoomConcurrencyTests.test_TC_ALLOC_02_TC_DATA_01_UT_ALLOC_05_two_threads_one_bed
IT_ALLOC_01	allocations.tests.AllocationCapacityIntegrationTests.test_TC_ALLOC_01_IT_ALLOC_01_active_allocation_reduces_room_capacity allocations.tests.AllocationViewTests.test_TC_ALLOC_01_IT_ALLOC_01_admin_allocates_through_view
IT_ALLOC_02	allocations.tests.AllocationViewTests.test_TC_ALLOC_02_IT_ALLOC_02_full_room_shows_error_and_saves_nothing
ST_ALLOC_01	allocations.tests.AllocationViewTests.test_TC_ALLOC_02_ST_ALLOC_01_fill_room_through_views_then_third_fails
UT_MOD_01	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_04_UT_MOD_01_modify_to_room_with_free_bed
UT_MOD_02	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_04_UT_MOD_02_modify_to_full_room_rejected
UT_CAN_01	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_05_UT_CAN_01_cancel_frees_bed_and_keeps_row
UT_HIS_01	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_06_UT_HIS_01_history_records_each_change_by_admin
IT_STAT_01	students.tests.StudentHomeCardTests.test_TC_ALLOC_03_IT_STAT_01_card_shows_room_details_and_awaiting_check_in allocations.tests.AllocationViewTests.test_TC_ALLOC_03_IT_STAT_01_student_home_shows_allocation_or_none
IT_HIS_01	allocations.tests.AllocationViewTests.test_TC_ALLOC_06_IT_HIS_01_history_filtered_by_student allocations.tests.AllocationViewTests.test_TC_ALLOC_06_IT_HIS_01_student_gets_403_on_history
ST_ALLOC_02	allocations.tests.AllocationViewTests.test_TC_ALLOC_04_TC_ALLOC_05_TC_ALLOC_06_ST_ALLOC_02_modify_cancel_history_flow
UT_CI_01	allocations.test_checkin_checkout.CheckInServiceTests.test_TC_CI_01_UT_CI_01_check_in_sets_date_and_history

Test Case ID	Automated test (module.Class.test)
UT_CI_02	allocations.test_checkin_checkout.CheckInServiceTests.test_TC_CI_01_UT_CI_02_check_in_twice_is_rejected
UT_CO_01	allocations.test_checkin_checkout.CheckOutServiceTests.test_TC_CO_01_UT_CO_01_check_out_completes_and_frees_bed
UT_CO_02	allocations.test_checkin_checkout.CheckOutServiceTests.test_TC_CO_01_UT_CO_02_check_out_before_check_in_is_rejected
IT_CO_01	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CO_01_IT_CO_01_check_out_through_url_updates_availability
IT_CO_02	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CO_01_IT_CO_02_student_cannot_check_in_or_out
ST_CICO_01	allocations.test_checkin_checkout.CheckInOutSystemTests.test_TC_CI_01_TC_CO_01_ST_CICO_01_full_stay
UT_CI_03	allocations.test_checkin_checkout.CheckInServiceTests.test_TC_CI_01_UT_CI_03_check_in_on_inactive_allocation_is_rejected
IT_CI_01	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CI_01_IT_CI_01_check_in_requires_post
UT_CO_03	allocations.test_checkin_checkout.CheckOutServiceTests.test_TC_CO_01_UT_CO_03_check_out_twice_is_rejected
IT_CO_03	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CO_01_IT_CO_03_check_out_get_shows_confirmation_only
UT_DASH_01	dashboard.tests.DashboardDesignedCaseTests.test_TC_DASH_01_UT_DASH_01_summary_counts_match_data
IT_DASH_01	dashboard.tests.DashboardDesignedCaseTests.test_TC_DASH_01_IT_DASH_01_dashboard_access_by_role
ST_DASH_01	dashboard.tests.DashboardDesignedCaseTests.test_TC_DASH_01_ST_DASH_01_displayed_values_match_database
ST_PERF_01	dashboard.tests.LoadTests.test_TC_PERF_01_ST_PERF_01_ninetieth_percentile_under_three_seconds
ST_REL_01	All records, the check-in and both history rows present and unchanged after restart
ST_SEC_01	accounts.tests.PasswordSecurityStorageTests.test_TC_SEC_03_ST_SEC_01_different_users_with_same_password_have_unique_hashes accounts.tests.PasswordSecurityStorageTests.test_TC_SEC_03_ST_SEC_01_passwords_stored_hashed_with_pbkdf2
ST_SEC_02	accounts.tests.RoleAuthorizationAdminProtectionTests.test_TC_SEC_02_IT_AUTH_03_ST_SEC_02_student_cannot_access_any_admin_page
ST_SEC_04	accounts.tests.InputSanitizationAndValidationTests.test_TC_SEC_04_ST_SEC_04_500_character_fuzzing_boundary accounts.tests.InputSanitizationAndValidationTests.test_TC_SEC_04_ST_SEC_04_csrf_token_required_on_post accounts.tests.InputSanitizationAndValidationTests.test_TC_SEC_04_ST_SEC_04_xss_payloads_rejected
ST_UX_01	Not yet executed: needs three students who did not build HMS; planned for the UAT demo
ST_MAINT_01	accounts.tests.ModularArchitectureMaintenanceTests.test_TC_MAINT_01_ST_MAINT_01_installed_apps_decoupled
ST_DATA_01	allocations.tests.CapacityConsistencySystemTests.test_TC_DATA_01_ST_DATA_01_random_actions_keep_capacity_consistent
IT_SEC_01	accounts.tests.EndToEndFuzzingTests.test_TC_SEC_04_IT_SEC_01_hostile_form_input_is_rejected_or_stored_as_text accounts.tests.EndToEndFuzzingTests.test_TC_SEC_04_IT_SEC_01_search_returns_no_extra_records_and_no_error

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IT_SEC_02	accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_server_error_page_has_no_stack_trace accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_student_on_admin_page_gets_custom_403 accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_unknown_record_gets_custom_404 accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_unknown_url_gets_custom_404
ST_CI_01	Workflow commands pass locally; confirm the green run on the GitHub Actions tab